New Kochen–Specker sets from algebraic number fields with enhanced contextual advantage

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We report two previously unknown Kochen–Specker (KS) sets in dimension 3 arising from algebraic number fields not previously explored in this context: the Heegner-7 ring $\mathbb{Z}[(1+\sqrt{-7})/2]$ yields a 43-vector set with a genuinely new orthogonality graph, and the golden ratio field $\mathbb{Q}(\varphi)$ yields a 52-vector set that is invisible to direct alphabet search and revealed only by cross-product completion. We show that the Heegner-7 ray pool exhibits the strongest Cabello–Severini–Winter contextual advantage ($\vartheta/\alpha = 1.215$) of any known algebraic KS construction in dimension 3. This advantage is driven by *auxiliary rays*—rays that contribute orthogonality edges without participating in any basis—and is entirely destroyed by cardinality minimization. The mechanism identifies a structural tension between minimizing a KS proof and maximizing its operational contextual strength.

INTRODUCTION

The Kochen–Specker (KS) theorem [?] proves that quantum observables cannot be assigned pre-existing values independently of measurement context. In dimension 3, all known KS constructions draw their ray coordinates from a handful of algebraic number fields: the integers [?], $\mathbb{Z}[\sqrt{2}]$ (the Peres construction [?]), and the Eisenstein integers $\mathbb{Z}[\omega]$ (Cabello's recent simplest KS set [?]). Whether other number fields support KS sets, and what operational consequences follow, has not been systematically investigated.

We report two new algebraic KS sets and identify a mechanism by which the algebraic origin controls the strength of the quantum contextual advantage in the Cabello–Severini–Winter (CSW) framework [?].

CONSTRUCTION METHOD

Given an alphabet $\mathcal{A} \subset \mathbb{C}$ (a finite set of allowed coordinates), we generate all rays $[v] \in \mathbb{C}P^2$ whose homogeneous coordinates lie in $\mathcal{A}$, identify orthogonal pairs via $\langle u|v\rangle = 0$, and collect orthonormal bases (triples of mutually orthogonal rays). KS-uncolorability—the nonexistence of a consistent $\{0,1\}$-valued coloring assigning exactly one 1 per basis—is decided by SAT encoding [?]. We minimize by randomized greedy deletion (500 trials, seed 42).

For fields where the basic alphabet yields too few orthogonalities, we apply *cross-product completion*: iteratively adding $u \times v$ (normalized) for all orthogonal pairs, then re-scanning for new orthogonalities and bases. This closes the ray pool under the geometric operations forced by the Hilbert space structure and can reveal KS sets invisible to the raw alphabet.

TWO NEW KS SETS

The Heegner-7 island. The imaginary quadratic field $\mathbb{Q}(\sqrt{-7})$ has class number 1, with ring of integers $\mathbb{Z}[\alpha]$ where $\alpha = (1+\sqrt{-7})/2$ satisfies $|\alpha|^2 = 2$. The alphabet $\{0, \pm1, \pm\alpha, \pm\bar{\alpha}\}$ generates 45 rays with 12 bases. Cross-product completion expands this to 145 rays and 42 bases. The completed pool is KS-uncolorable, with a minimum of **43 vectors in 23 bases**.

This 43-vector set has a genuinely new orthogonality graph: it is not isomorphic to any previously catalogued KS set. Its degree sequence, spectral properties, and Weisfeiler–Leman color refinement distinguish it from all known constructions.

The golden ratio island. The golden ratio $\varphi = (1+\sqrt{5})/2$ satisfies $\varphi^2 = \varphi + 1$, giving $|\varphi|^2 \approx 2.618$—above the norm-2 threshold that controls KS-uncolorability in quadratic fields. The basic alphabet $\{0, \pm1, \pm\varphi\}$ is colorable. However, cross-product completion generates rays with coordinates in the full ring $\mathbb{Z}[\varphi]$, including $\varphi - 1 = 1/\varphi$, and the completed pool of 193 rays is KS-uncolorable with a minimum of **52 vectors**.

This construction is invisible to standard alphabet searches: the golden ratio island is revealed only by the geometric closure operation, not by the algebraic coordinates alone.

To our knowledge, neither the Heegner-7 nor the golden ratio KS set has been previously reported.

CONTEXTUAL ADVANTAGE

The CSW framework [?] maps the orthogonality graph G of a ray set to a noncontextuality inequality via three invariants:

$$\alpha(G) \le \vartheta(G) \le \alpha^*(G),$$

where α is the independence number (classical bound), ϑ is the Lovász theta number (quantum bound), and α^*

TABLE I. CSW invariants for algebraic KS islands. Top: minimized sets. Bottom: full ray pools. The ratio ϑ/α measures contextual advantage.

Minimized KS sets				
Island	n	α	ϑ	ϑ/α
Integer (CK-31)	31	11	11.71	1.065
Eisenstein	33	12	12.05	1.004
Peres	33	12	12.00	1.000
$\mathbb{Z}[\sqrt{-2}]$	33	12	12.00	1.000
Heegner-7	43	16	16.00	1.000
Full ray pools				
Island	n	α	ϑ	ϑ/α
Heegner-7	145	46	55.89	**1.215**
Eisenstein	57	18	19.34	1.074
Integer	49	17	17.70	1.041
Peres	49	23	23.00	1.000
$\mathbb{Z}[\sqrt{-2}]$	49	23	23.00	1.000

is the fractional packing number. A quantum advantage exists when $\vartheta > \alpha$.

Table **??** compares the CSW invariants for minimized KS sets and full ray pools across the known algebraic islands.

The Heegner-7 full pool achieves $\vartheta/\alpha = 1.215$—a 22% quantum advantage, the strongest of any known algebraic KS construction in dimension 3. Yet its minimized 43-vector set is ϑ-perfect ($\vartheta = \alpha$). This dramatic reversal demands explanation.

THE AUXILIARY RAY MECHANISM

We identify the structural origin of this effect. Define a ray as *basis-participating* if it belongs to at least one orthonormal basis, and *auxiliary* otherwise. Auxiliary rays have orthogonal neighbors in the graph but do not appear in any complete triple.

KS-uncolorability depends on the basis *hypergraph*: auxiliary rays are invisible to it. But the CSW framework depends on the orthogonality *graph*: auxiliary rays contribute edges that constrain the independence number α without adding bases.

In the Heegner-7 pool, 76 of 145 rays (52%) are auxiliary. These rays add 522 orthogonal pairs, including 9 hub vertices of degree 16, severely restricting independent sets ($\alpha/n = 0.317$). The SDP relaxation exploits these constraints, distributing weight fractionally to achieve $\vartheta/n = 0.385$.

When minimization reduces the pool to 43 vectors, it removes auxiliary rays first—they are dispensable for uncolorability. But these are precisely the rays that enabled $\vartheta > \alpha$. The contextual advantage is destroyed.

The Hoffman bound provides a spectral explanation (Table **??**). Pools with $\alpha/H < 1$ (spectrally tight) exhibit CSW violations; pools with $\alpha/H > 1$ are ϑ-perfect. The Peres and $\mathbb{Z}[\sqrt{-2}]$ pools, despite having 33% auxiliary rays, remain ϑ-perfect because their orthogonality graphs are disconnected: auxiliary rays form loosely connected components that do not bridge the main structure, and ϑ decomposes additively over components [**?**].

TABLE II. Auxiliary ray fraction and spectral tightness. H = Hoffman bound on α. Spectral tightness $\alpha/H < 1$ correlates with CSW violation.

Island	Aux. frac.	α/H	ϑ/α
Heegner-7	52%	0.898	**1.215**
Eisenstein	0%	0.904	1.074
Integer	0%	0.977	1.041
Peres	33%	1.176	1.000
$\mathbb{Z}[\sqrt{-2}]$	33%	1.176	1.000

DISCUSSION

Our results reveal a structural tension at the heart of KS contextuality: *minimizing the proof size and maximizing the contextual strength are competing objectives.* The rays that are dispensable for the logical proof of contextuality (the auxiliary rays) are precisely those that amplify its operational strength in the CSW framework. This suggests that the standard practice of seeking minimal KS sets, while natural from a proof-complexity perspective, discards operationally relevant structure.

The Heegner-7 island's 22% advantage is modest in absolute terms but structurally significant: it arises from the interplay between the algebraic number field (which determines the auxiliary ray density) and the spectral properties of the resulting orthogonality graph (which determine whether auxiliary rays translate into contextual advantage). The golden ratio island, with its cross-product completion mechanism, demonstrates that KS sets can hide in algebraic fields whose basic coordinates are insufficient—geometric closure can unlock contextuality where algebra alone cannot.

Both constructions arise from the same underlying principle: the generator norm $|\alpha|^2 \leq 2$ controls KS-uncolorability across all tested number fields. The integer identity $1+1 = 2$, the Eisenstein identity $1+\omega+\omega^2 = 0$, and the Heegner-7 identity $\alpha\bar{\alpha} = 2$ are different manifestations of this single constraint. A companion paper [**?**] presents the full algebraic classification.

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